

编译LVGL工程

获取工程

- `lvgl_100ask_course_v9\part3\02_codes\lv_port_linux`
- https://github.com/lvgl/lv_port_linux

编译

- Makefile: 阅读[README.md](#)
 - 如果出错，在 `lv_conf.h` 中关闭宏：`LV_USE_LOTTIE`、`LV_USE_THORVG_INTERNAL`、`LV_USE_VECTOR_GRAPHIC`
- cmake: 阅读[README.md](#)

交叉编译

Makefile

修改Makefile (`lvgl_100ask_course_v9\part3\02_codes\lv_prj_new_cmake_and_Makefile\Makefile`)

```
#
# Makefile
#
#命令行执行下面三行
#export PATH=$PATH:/home/ubuntu/100ask/T113s4_Lvgl9-Class_TinaSDK-
V1/prebuilt/rootfsbuilt/arm/toolchain-sunxi-glibc-gcc-830/toolchain/bin
#export CROSS_COMPILE=arm-openwrt-linux-gnueabi
#export STAGING_DIR=/home/ubuntu/100ask/T113s4_Lvgl9-Class_TinaSDK-
V1/out/t113_s4/100ask_lvgl9/openwrt/staging_dir/target

#CC           ?= gcc
#CXX          ?= g++
CC            = arm-openwrt-linux-gnueabi-gcc
CXX           = arm-openwrt-linux-gnueabi-g++
LVGL_DIR_NAME ?= lvgl
LVGL_DIR      ?= .

WARNINGS      := -Wall -Wshadow -Wundef -Wmissing-prototypes -Wno-discarded-qualifiers -
Wextra -Wno-unused-function -Wno-error=strict-prototypes -Wpointer-arith \
               -fno-strict-aliasing -Wno-error=cpp -Wuninitialized -Wmaybe-uninitialized
               -Wno-unused-parameter -Wno-missing-field-initializers -Wtype-limits \
               -Wsizeof-pointer-memaccess -Wno-format-nonliteral -Wno-cast-qual -
Wunreachable-code -Wno-switch-default -Wreturn-type -Wmultichar -Wformat-security \
```

```

        -Wno-ignored-qualifiers -Wno-error=pedantic -Wno-sign-compare -Wno-
error=missing-prototypes -Wdouble-promotion -Wclobbered -Wdeprecated -Wempty-body \
        -Wshift-negative-value -Wstack-usage=2048 -Wno-unused-value -std=gnu99
CFLAGS      ?= -O3 -g0 -I$(LVGL_DIR)/ $(WARNINGS)
LDFLAGS     ?= -lm -lstdc++
BIN         = main
BUILD_DIR   = ./build
BUILD_OBJ_DIR = $(BUILD_DIR)/obj
BUILD_BIN_DIR = $(BUILD_DIR)/bin

prefix      ?= /usr
bindir      ?= $(prefix)/bin

#Collect the files to compile
MAINSRC     = ./main.c

include $(LVGL_DIR)/lvgl/lvgl.mk

CSRCS       += $(LVGL_DIR)/mouse_cursor_icon.c

OBJEXT      ?= .o

AOBJS       = $(ASRCS:.S=$(OBJEXT))
COBJS       = $(CSRCS:.c=$(OBJEXT))
CXXOBJS     = $(CXXSRCS:.cpp=$(OBJEXT))

MAINOBJ     = $(MAINSRC:.c=$(OBJEXT))

SRCS        = $(ASRCS) $(CSRCS) $(CXXSRCS) $(MAINSRC)
OBJS        = $(AOBJS) $(COBJS) $(CXXOBJS) $(MAINOBJ)
TARGET      = $(addprefix $(BUILD_OBJ_DIR)/, $(patsubst ./%, %, $(OBJS)))

## MAINOBJ -> OBJFILES

all: default

$(BUILD_OBJ_DIR)/%.o: %.cpp
    @mkdir -p $(dir $@)
    @$(CXX) $(CXXFLAGS) -c $< -o $@
    @echo "CXX $<"

$(BUILD_OBJ_DIR)/%.o: %.c
    @mkdir -p $(dir $@)
    @$(CC) $(CFLAGS) -c $< -o $@
    @echo "CC $<"

$(BUILD_OBJ_DIR)/%.o: %.S
    @mkdir -p $(dir $@)
    @$(CC) $(CFLAGS) -c $< -o $@
    @echo "CC $<"

default: $(TARGET)
    @mkdir -p $(dir $(BUILD_BIN_DIR)/)
    $(CC) -o $(BUILD_BIN_DIR)/$(BIN) $(TARGET) $(LDFLAGS)

```

```
clean:
    rm -rf $(BUILD_DIR)

install:
    install -d $(DESTDIR)$(bindir)
    install $(BUILD_BIN_DIR)/$(BIN) $(DESTDIR)$(bindir)

uninstall:
    $(RM) -r $(addprefix $(DESTDIR)$(bindir)/,$(BIN))
```

Makefile commit: https://github.com/lvgl/lv_port_linux/commit/ee4f35247dbcd0fa4c94c32ad822943318e906a1

编译命令

```
make -j
```

cmake

添加文件:

```
lvgl_100ask_course_v9\part3\02_codes\lv_prj_new_cmake_and_Makefile\user_cross_compile_setup.cmake
```

user_cross_compile_setup.cmake已提交PR: https://github.com/lvgl/lv_port_linux/pull/70

编译命令:

```
cmake -DCMAKE_TOOLCHAIN_FILE=./user_cross_compile_setup.cmake -B build -S .
make -C build -j
```

技术交流学习

欢迎加入讨论:

- 社区交流: <https://forums.100ask.net>
- QQ技术交流群 (如群满, 请加qq: 401684796 验证备注: LVGL): 962138234
- 微信交流群: 添加微信: baiwenkeji_fae 验证备注: LVGL